	Supply V (V)	1					
	activity factor	0.15					
	Frequency	800MHz					
	Case	Input trans. time	Load Cap.	Peak short-circ. current	short-circ pulse width		
	A	20	20	90	18		
	B	80	20	210	70		
	C	80	200	95	55		
		(ps)	(fF)	(microA)	(ps)		
2A	short circuit energy per transition:						
	P = IV *t						
	however, since the pulses are triangular, area of triangle = 0.5*h*w						
	therefore:						
	P = 0.5* IV*T						
	Case	Peak short-circ. current	short-circ pulse width	**Short Circuit Energy per transition**			
	A	90	18	**810**	**0.81fJ**		
	B	210	70	**7350**	**7.35 fJ**		
	C	95	55	**2612.5**	**2.6125fJ**		
		(microA)	(ps)				
	Average short circuit power:			Note: shape triangle.			
	1/2 * i*v *t*f						
2B	Case	Peak short-circ. current	Voltage V	T time ps	F	Avg. Short Circuit Power	**Avg. Short Circuit Power**
	A	90	1	18	800	648000	**648nW**
	B	210	1	70	800	5880000	**5880nW**
	C	95	1	55	800	2090000	**2090nW**
		(microA)		(ps)	MHz		
	Case	Input trans. time	Load Cap.	short-circ pulse width			
	B	80	20	70			
	C	80	200	55			
		(ps)	(fF)	(ps)			
2C	B and C have the same input transition time but different load capacitance. A larger load can reduce the Short Circuit Current Pulse time, as it requires more to actually change value and output the stored charge and so the output value is closer to Vdd or 0. I tend to imagine it like water from a fire hose? It requires a lot of energy to turn on, so the circuit spends less time with both NMOS and PMOS passing energy and a lot more time closer to the 'ideal' values.						
	This happens even if capacitive switching energy increases because that's a whole different mechanism. Switching energy is related to the amount of energy needed to switch the state and is only related to Cl and Vdd. it isn't related to the current pulse time at all.						

I apologise for the trouble! Due to my disability, my wrists and hands are not working well at the moment so I must type all my work. I hope that I have formatted my work in a clear enough manner to mark, but if not, please let me know and I will explain my reasonings over a Zoom call or in person.



	Case	Input trans. time	Load Cap.	Peak short-circ. current	short-circ pulse width		
	A	20	20	90	18		
	C	80	200	95	55		
		(ps)	(fF)	(microA)	(ps)		
	Supply V (V)	1					
	activity factor	0.15					
	Frequency	800MHz					
	E (low to high)	equals	Cl Vdd^2				
2D	Case	Load Cap.	Supply Voltage	**E Low to High**			
	A	20	1	**20**			
	C	200	1	**200**			
		(fF)		**(fJ)**			
	Case	Short Circuit Energy per transition					
	A	0.81fJ					
	C	2.6125fJ					
	In both cases A and C, the short circuit energy per transition is far less than the capacitive switching energy. Case C's short circuit energy is comparatively much smaller, proportional to the ordinary capacitive switching energy, than Case A.						
	Option 1:						
	Reduce the current flowing when the circuit is switching values and short circuits. This would cause lower current values in the circuit overall, though, which would not be good because the circuit's speed would dramatically decrease. This could also be done by lowering Vdd to double Vth, to lower the voltage that surpasses the threshold voltage.						
2E							
	Option 2:						
	Change the gates! Use transistors with a high Vth value to reduce the time that short circuiting could occur. A drawback for this is that values take longer to propagate through the circuit which also is not good.						